

UNIT TEST REPORT

By Firmgen

Report identity

Project	espblynk
Target	firmware (verified off target, no MCU hardware in the loop)
Report ID	UTR-7ED2C62B
Run ID	7ed2c62b-945c-42d2-b7a6-c6d77c5e03d4
Execution date	15 October 2026, 07:18 UTC
Prepared by	Firmgen automated unit-test agent
Cases executed	0 across 0 test file(s)

1. Test Objective

This activity establishes no executed unit-test evidence: 0 of 0 cases passed, 0 of 0 requirements are traced as passing, and 0 of 29 enumerated firmware units are verified. It supports an assessment of the current evidence status only; it does not establish whether the firmware behavior is correct or incorrect.

2. Environment and Scope

Source under test	espblynk
Compile-time defines	UNIT_TEST, HOST_TEST, UNITY_INCLUDE_EXEC_TIME
Test parallelism	8 thread(s)
Test framework	Ceedling 1.1.6 (Unity + CMock)
Build system	Ceedling 1.1.6
Compiler	C:\Users\Adith\AppData\Local\craftifai\toolchains\mingw\16.2.0posix-14.0.0-ucrt-r1\bin\gcc.exe 16.2.0
Host environment	FirmGen-managed toolchain (ruby 3.3.12-1) on Windows-10
Execution mode	Off target - compiled for the build architecture, no MCU hardware in the loop; executed directly on the build host, not in a container
Test cases executed	0 across 0 test file(s)

3. Test Harness and Reproducibility

The measured environment specifies Ceedling 1.1.6 with Unity and CMock, built with the UNIT_TEST, HOST_TEST, and UNITY_INCLUDE_EXEC_TIME defines, using the recorded GCC 16.2.0 toolchain on Windows 10. Execution was configured off target on the build host with 8-thread parallelism, but no test files were executed; consequently, there is no executed evidence of mocks, fakes, or stubs used.

Reproduction step	Command	Result
Configure	`ceedling project.yml (generated per run)`	● PASS
Build and link	`ceedling test:build_only`	● PASS
Execute tests	`ceedling test:all`	● PASS

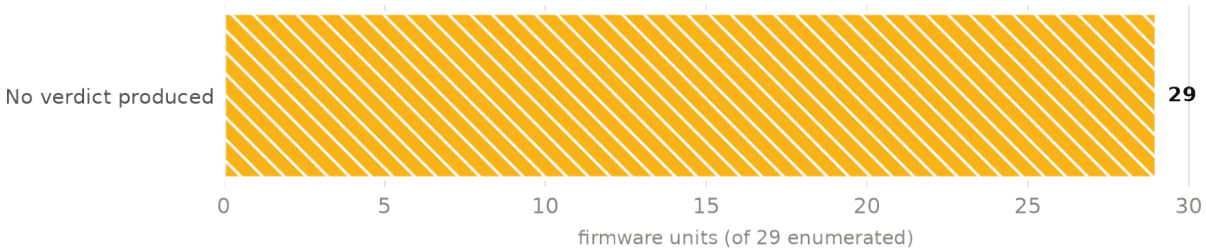
Evidence shipped with this run: executed_tests.xlsx, run.txt

4. Execution Summary

0	0%	0	0
Cases passed	Case pass rate	Cases executed	To address

Metric	Result
Cases passed	0
Case pass rate	0.0%
Requirements fully passing	0
Requirements traced to test evidence	0
Total test cases	0
Test files executed	0
Requirement-driven cases	0
Firmware-coverage cases	0
Cases awaiting a verdict	0
Cases with a failure to address	0
Requirements awaiting a verdict	0
Requirements with a failure to address	0
Total execution time	0 ms

Firmware units verified by this suite



5. Traceability

No requirement cases were executed, and 0 of 0 requirements are traced as passing; the totals also report no requirements not yet verified or to address. This means there is no requirement-level evidence in this activity, not that requirements have passed or failed.

Requirements traced to executed cases

No requirement-driven cases were executed in this run.

Firmware units traced to executed cases

No firmware-coverage cases were executed in this run.

6. Failure, Ignored Case, and Root-Cause Record

Nothing outstanding: every executed case asserted a value and got it, and no case was left without a verdict.

7. Limitations and Exclusions

This report states what the executed cases established. The boundaries below mark where off-target unit testing ends and other verification activities begin - they are the scope of this activity, not defects in it.

- This activity does not establish the behavior of any of the 29 enumerated firmware units.
- Host unit tests prove nothing about electrical behaviour, interrupt latency or RTOS scheduling.
- No MCU hardware was in the loop, so this activity does not establish behavior on target hardware.
- With no test files or coverage cases executed, this activity does not establish code coverage.

8. Verification Status and Conclusion

Decision field	Conclusion
Verified scope	0 of 0 executed case(s) passed, covering 0 requirement(s) in full across 0 test file(s).
Outstanding items	Determine why no test files were executed, then run the intended unit-test suite and record its results and requirement traceability for the enumerated firmware units.
Conclusion	The authoritative totals show no executed cases or passing requirement traces, while all 29 enumerated units remain without a verdict. The evidence supports reporting the current verification status and identifying execution of the unit-test suite as the next step; it does not support a conclusion about firmware correctness or a release decision.

Generated automatically from the executed test results of run 7ed2c62b-945c-42d2-b7a6-c6d77c5e03d4. The figures in this document are computed from executed_tests.xlsx; no count in it was written by a language model.

9. Appendix: Complete Case Record

This run executed no test cases, so there is nothing to record here. Section 2 states what was read off disk.